

**Comparative Analysis of ML classification models
Applied to Deepfake Media Detection Dataset**

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Kaggle dataset: **Deepfake & Synthetic Media Detection Dataset**

Abstract

The dangerous proliferation of artificially produced media claiming to be authentic, or better known as deepfakes, poses substantial threats to information integrity, democratic discourse, and individual privacy, as the technologies used to produce these deepfakes have become increasingly sophisticated and efficient. Generative adversarial networks (GANs), diffusion models, face-swapping, and voice cloning technologies have collectively lowered the barrier to producing convincing artificial content to the point that the human eye is no longer a reliable means of examination. Automated, scalable detection systems are therefore a pressing research and engineering priority.

This paper details a comparative study of four supervised machine learning classification models: logistic regression, support vector machine, random forest, and gradient boosting. These algorithms were applied to a metadataset with deepfake detection data, sourced from Kaggle. The dataset is made up of 1000 media samples that are measured by nine features that include: lip-synchronisation scores, visual artefact scores, lighting inconsistency scores, and compression levels. A key preliminary step was identifying and removing a data leakage source, the `generation_method` column. In this column, every real source has “none”, while every fake source has a value: GAN, FaceSwap, Diffusion, VoiceClone. This means that the target label would have been encoded, and all subsequent evaluation would be invalidated. Exploratory Data Analysis revealed that three features have zero overlap value between real and synthetic classes, meaning the classification of this dataset would be easily separable.

The classifiers used were logistic regression, random forest, gradient boosting, and support vector machine. They were trained on an 80/20 train/test split and were evaluated by Accuracy, precision, recall, and F-1 score.

1. Introduction

Deepfake is the term used to describe the manipulation of already existing media or the generation of synthetic media using deep learning techniques, this is usually in the form of fake videos, fake images, and fake audio (Altuncu et al., 2024). The growth of deepfakes creates serious risks surrounding misinformation, fraud, and privacy violations and has led to a growing scepticism surrounding the authenticity of media content (Amerini et al., 2025). These techniques have advanced to the point where Masood et al. (2022) found that manipulations of speech and audio were virtually indistinguishable to humans. This means that human inspection is now an unreliable defence.

Existing research has explored detection across several approaches, such as analysing raw pixel data, examining frequency domain fingerprints, and checking audio-visual synchronisation. However, sources such as Masood et al. (2022) note that these techniques generalise quite poorly. Detection performance has been found to drop significantly when models are tested on deepfakes generated by methods different from those they were trained on.

This study investigates whether conventional machine learning classifiers, trained on a structured dataset, can reliably distinguish between real and fake media. Four common classification models are benchmarked and compared.

2. Literature Review

There is an extensive review of deepfake detection in the academic literature. Given the advancement of the deepfake generation technology itself, it is fast-moving and covers a wide range of methodological approaches. A critical reading of the most recent literature reveals both considerable methodological progress and persistent, unresolved challenges that directly motivate the metadata-based classification approach taken in this study.

Gong and Li (2024) provide a broad overview of facial forgery detection research from 2019 to 2023, tracing how the field has developed alongside advances in generative technology. They outline a clear progression from early GAN-based approaches, which, due to their visible colour inconsistencies and unnatural facial features, were relatively easy to tell apart. Through to the more recent diffusion models, which are substantially more difficult to detect and increasingly hard for humans to identify. They organise current detection approaches into four main categories: standard CNN-based methods, CNNs incorporating semi-supervised learning, transformer-based architectures, and methods that make use of biological signals such as pulse rate and blinking patterns. A key finding from their research is that when models were tested on datasets different to what they were trained on, detection accuracy drops significantly across all four categories when models are exposed to previously unseen manipulation types. It indicates that strong dataset performance is not a reliable predictor of how well a model will generalise in practice. This finding is further supported by other studies mentioned below.

Gupta et al. (2023) offer a broad view of deepfake detection research over the period 2015-2023, examining 67 studies across both image and video detection and speaker authentication. A finding of their research is that fusion-based approaches, those that draw on multiple media modalities simultaneously, can outperform methods that rely on a single modality. Specifically, experimental results demonstrate that joint audio-visual deepfake detection outperforms audio-only or visual-only approaches. These findings are directly relevant to the feature set used in this study, which combines signals from three distinct dimensions: the `lip_sync_score` measures coherence between audio and visual alignment, the `visual_artefacts_score` captures forensic irregularities at the image level, and the `lighting_inconsistency_score` assesses inconsistencies in spatial and physical lighting.

Wang et al. (2024) offer a different approach to evaluating deepfake detection systems by looking at not just accuracy but also the reliability of the detection systems in broader real-world contexts. They highlight three areas that are often overlooked in the existing literature: whether models can transfer effectively across different datasets, whether their decisions can be understood and explained, and whether they hold up under realistic, noisy conditions. Their central argument is that strong benchmark performance won't necessarily translate to model reliability when deployed outside of controlled environments. This finding is reflected in the results of this report; the very high accuracy score achieved by our classification models is a reflection of the clean and controlled nature of the Kaggle dataset

rather than evidence of robust model performance.

A survey of deepfake generation and detection by Liu et al. (2025) traces how synthetic media creation evolved from early GAN-based models, where a generator and discriminator worked in opposition to each other to produce increasingly realistic outputs, through to diffusion models that work on a different basis and have become an increasingly prominent approach for high-quality synthetic face generation. They note that while GANs were transformative, they are limited by issues such as training instability and a tendency to produce less varied outputs. Diffusion models have largely addressed these shortcomings, and as a result, they generate content that is considerably harder to detect. Diffusion is one of the four generation methods in the dataset. Despite the success in classifying these deepfakes in our study, in reality, Liu et al. describe the relationship between generation and detection as a “cat and mouse game”; it is an ongoing and evolving dynamic rather than a solved problem.

3. Methodology

3.1 Dataset Description

The dataset used in this study is the Deepfake and synthetic media detection dataset from Kaggle. It contains 1000 media records, 501 of which are labelled as real and 499 as fake. This makes it an almost perfectly balanced binary classification problem. Each record represents a single media item (image, audio, or video) and contains 11 descriptive attributes. The continuous numerical features are lip_sync_score, visual_artifacts_score, compression_level, and lighting_inconsistency_score. One feature is a discrete integer count: face_count, ranging from 0 to 4. And the remaining features are categorical strings: media_type, content_category, audio_present, and source_platform.

3.2 Exploratory Data Analysis

Before any models were built, a thorough exploratory analysis was carried out to identify any issues with the data. In this case, two issues were identified.

The first issue identified was a data leakage issue. The column generation_method contains information about how each piece of fake data was created, either by GAN, diffusion, Faceswap, or voiceclone. Every real piece of data had no record in this column, while every fake sample did. This means a model could classify: real or fake, simply by checking to see if this column contained a value for each specific instance. This would mean the model would just read the dataset and know the answer without learning anything meaningful about what makes the media real or fake. This column was therefore removed before model training began.

The second issue identified was that three of the features: lip_sync_score, visual_artifacts_score, and lighting_inconsistency_score were so clearly divided between real and fake that there was no grey area whatsoever. For artefacts and lighting measures, real data always scored below a certain threshold and fake always scored above. The opposite was the case for lip synchronisation, fake always scored low and real scored high.

The gap between real and fake was absolute with no overlap.

This is essentially the reason why model performance appeared to be so accurate. The models were able to determine based on a simple threshold rule.

The remaining features provide almost no useful information, compression level showed statistically indistinguishable means for real and fake. Face count and all of the categorical features split at approximately 50/50 real vs. fake regardless of value, meaning knowing any of them told the model nothing about whether a sample was genuine or not.

3.3 Preprocessing

Following the exploratory analysis and the removal of the generation_method column, the data was prepared for model training. Categorical features were ordinally encoded, converting them into integer values; the target variable was encoded as 1 (Fake) and 0 (Real). The dataset was partitioned into 80% train and 20% test using stratified random sampling, using a fixed random seed of 42 to ensure class distribution was preserved across all splits.

Standardisation was applied to logistic regression and support vector machine, as they are sensitive to differences in the scale of features. Standardisation transforms their mean to 0 and standard deviation to 1. The standardisation parameters were calculated using the training data, then applied to the test data.

3.4 Classification Algorithms

Four ML algorithms were chosen for this study to give a meaningful comparison rather than just report results from a single approach. The four models are logistic regression, random forest, gradient boosting, and support vector machine.

3.5 Evaluation

Models were evaluated using standard binary classification metrics on the held-out test set. Accuracy measures the model's overall classification correctness. Precision measures the proportion of positive predictions that are actually positive; this captures the false alarm rate. Recall measures the proportion of actual positives that were correctly identified, capturing the missed detection rate. The F-1 score is the harmonic mean of precision and recall; it provides a balanced single-number summary of the two scores. Five-fold stratified cross-validation was carried out on the training data to assess stability across different splits.

4. Results

4.1 Model Performance

The table below shows how each of the classification algorithms performed on the test data.

Model	Accuracy (%)	Precision (%)	Recall (%)	F-1 (%)
Logistic Regression	100	100	100	100
SVM	99.50	100	99	99.50
Gradient Booster	100	100	100	100
Random Forest	100	100	100	100

(Table 1: Classification model performance on test set)

Logistic Regression, Gradient Booster, and Random Forest all achieved perfect scores across all of the evaluation metrics. SVM achieved slightly lower with 99.5% accuracy and F-1 score, and 99% recall; however, the distinction is very minor as it only classified one wrong instance.

4.2 Feature Importance

A useful property of a random forest algorithm is that it can tell us how much each feature contributes to its predictions. This is done through mean decrease impurity. Features that were used more often and effectively to separate real from fake samples receive a higher score; the results of this are shown in the table below.

Feature	Importance (%)	EDA Findings
lighting_inconsistency_score	38.82	Real: 0.00–0.30 / Fake: 0.40–1.00: no overlap
visual_artifacts_score	31.80	Real: 0.00–0.30 / Fake: 0.40–1.00: no overlap
lip_sync_score	28.99	Real: 0.70–1.00 / Fake: 0.00–0.70: no overlap
source_platform	0.10	50/50 split across all platforms
compression_level	0.10	Identical means (0.52) for both classes
face_count	0.06	50/50 split at every count value

content_category	0.06	50/50 split across all categories
media_type	0.05	50/50 split across all types
audio_present	0.03	50/50 split for Yes and No

(Table 2: Feature Importance for Random Forest model predictions)

The results show us that three features account for over 99% of the model's predictions. These results are consistent with what was discovered during EDA; the features that the model identified as the most important were those that were found to have no overlap between the classes. This result also highlights that, despite being included in the dataset, the other features aren't genuinely useful for this classification task. They imply that a detection model could operate using only three features and lose almost nothing in terms of accuracy; however, this is completely reflective of the structured nature of the dataset and not of real-world complexities.

5. Discussion

Two aspects of this study require examination: firstly, the near-perfect scores achieved by all the classification models; and secondly, the nature of the dataset that produced these results, rather than the sophistication of the classifiers.

There is a clear alignment between the findings of the exploratory data analysis and the results produced by the models. Three features: lip_sync_score, visual_artifacts_score, and lighting_inconsistency_score can already separate real and fake samples by applying a simple manual rule to the data. The fact that more complex machine learning models can then produce the same results is therefore expected.

A systematic review of deepfake detection techniques by Sharma et al. (2024) demonstrates that advancements in machine learning techniques have made deepfakes increasingly realistic and harder to distinguish, but they note a particular limitation of current detection research, which is generalisability across models. This is very relevant to our results; the high accuracy does not reflect real-world conditions. These are the results of a clean, balanced, synthetic dataset with artificially sharp boundaries. In a real-world environment where complexities come into play, we would see the generalisation limitations, and the results would not be so impressive.

The results of the model's classification entirely reflect the engineered features of the dataset and not the power of the classification models themselves.

6. Conclusion

This research report presented a comparative analysis of four supervised machine learning algorithms applied to a deepfake detection dataset. Three of four models achieved perfect classification performance, with one, SVM, falling just short.

Despite this, the central finding of the report is not the model evaluation scores themselves but the understanding of why they occurred. Exploratory data analysis revealed that three features exhibit no overlap between real and fake classes meaning that the data was already separable before applying any algorithms to it. Furthermore, the importance of this step was demonstrated by the identification and removal of a data leakage issue through the `generation_method` column, which gave away the classification.

The results should therefore be understood in the context of the dataset's controlled, synthetic construction, and not evidence of robust classification algorithms. This is repeatedly demonstrated through the literature that clean model performance on test data will not directly translate to real-world examples. The model's ability to generalise is a current limitation of deepfake detection research that requires further work.

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